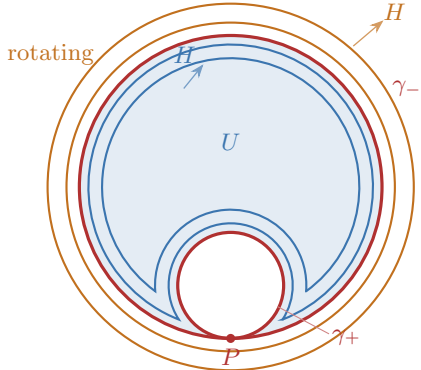
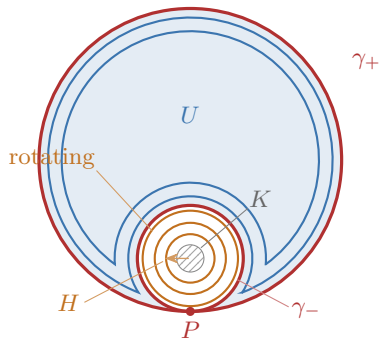




(a) rotating family outside γ_- :
every rotating loop encloses all of U ,
and its area grows from $\text{area}(\gamma_-)$.
The layout of the learned chart.



(b) the branches exchanged: rotating
family inside γ_- , nesting *inward*,
its area falling to $\text{area}(K) \geq 0$.